

Semantic Retrieval Transfer for Structure-Compatible Generalization

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Phase 5: frozen-encoder semantic transfer

Abstract

This phase moves structure-compatible generalization from rendered templates to short semantic retrieval cases. Actual frozen sentence encoders provide embeddings; nearest-neighbor structure infers candidate paraphrase and entity-substitution orbits; trained retrieval selectors are evaluated on held-out semantic variants without OOD labels.

1. Manifest

Field	Value
GPU	L4
encoders	['all_minilm_l6_v2', 'bge_small_en_v1_5']
configs per encoder	32
rows	66

Table 1. Semantic retrieval Modal L4 manifest.

2. Predictor Correlations

Domain	Predictor	Pearson r	Spearman r	N
semantic_retrieval_frozen_encoder	compatibility_true	0.940	0.915	64
semantic_retrieval_frozen_encoder	compatibility_discovered	0.861	0.863	64
semantic_retrieval_frozen_encoder	train_accuracy	0.772	0.628	64
semantic_retrieval_frozen_encoder	id_validation_accuracy	0.709	0.488	64
semantic_retrieval_frozen_encoder	compatibility_wrong	-0.872	-0.919	64

Table 2. Semantic retrieval predictors against OOD.

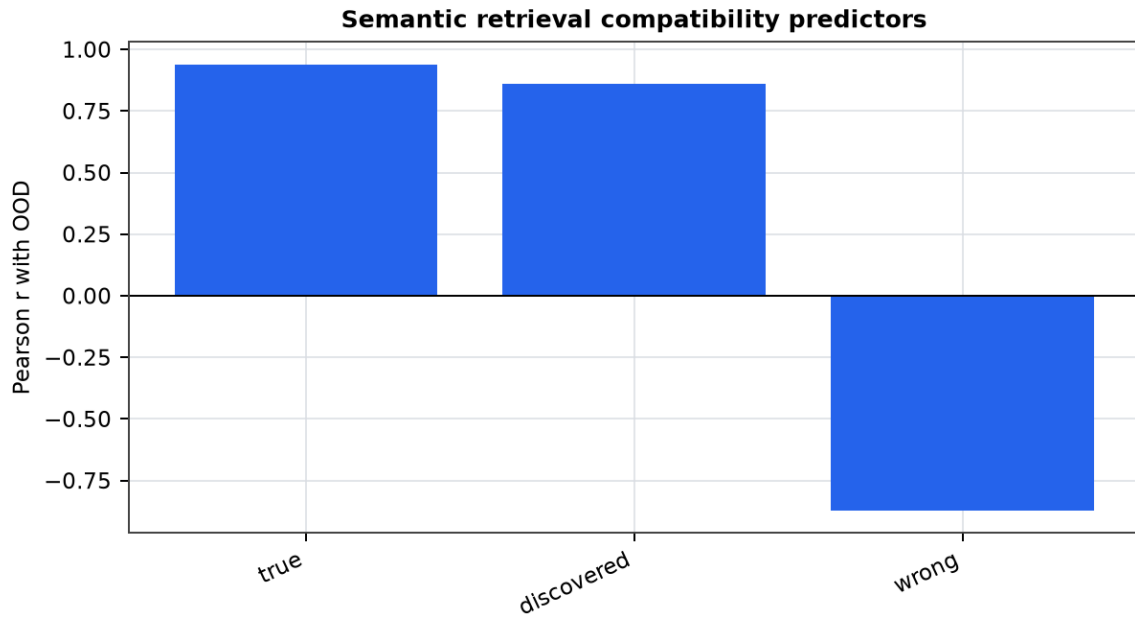


Figure 1. Semantic retrieval compatibility predictors.

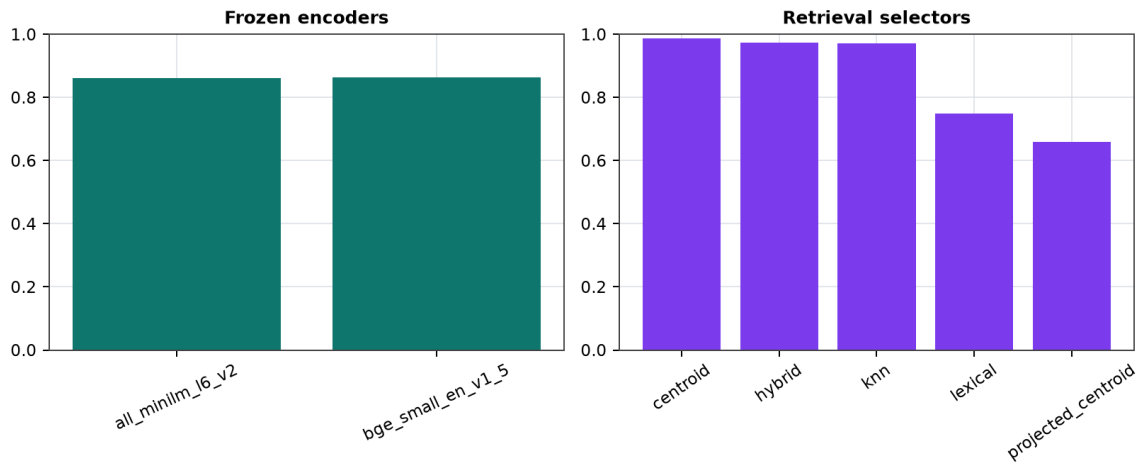


Figure 2. Encoder and selector breakdowns.

3. Breakdowns

Encoder	N	Mean train	Mean ID	Mean OOD	Compat
all_minilm_l6_v2	32	0.939	0.912	0.860	0.940
bge_small_en_v1_5	32	0.955	0.920	0.863	0.907

Table 3. Frozen encoder rows.

Family	N	Mean train	Mean ID	Mean OOD	Compat
centroid	13	1.000	1.000	0.988	0.985
hybrid	12	1.000	1.000	0.974	0.983
knn	12	0.969	0.911	0.971	0.966
lexical	13	1.000	1.000	0.750	0.870
projected_centroid	14	0.786	0.692	0.658	0.828

Table 4. Retrieval selector families.

4. Scope

The result is bounded to a finite semantic-retrieval corpus and public frozen text encoders. It does not certify arbitrary paraphrase invariance or production model behavior.